Neha Rao

Baltimore, MD · Remote-Ready · U.S. Authorized · raoneha0924@gmail.com · (781) 658-8282 · linkedin.com/in/nehabr · Portfolio

Professional Summary

Computational biologist and data scientist with an **M.S. in Bioinformatics (Boston University)** and 3+ years of experience spanning **genomic data analysis, machine learning, bioinformatics pipeline development, database engineering, and scientific data curation**. Co-author on two peer-reviewed publications. Proficient in **Python, R, SQL, and Bash** on Linux/Unix HPC; experienced with NGS tooling (Bowtie2, BWA, STAR, SAMtools, MACS2, DESeq2, Seurat, Scanpy), ML frameworks (XGBoost, PyTorch, scikit-learn), and full-stack data platform development (RShiny, SQL, JavaScript). Comfortable working at the interface of research science, data engineering, and cross-functional stakeholder communication.

Core Competencies

- **Genomics & Multi-Omics:** NGS pipeline development · bulk RNA-seq · scRNA-seq · snATAC-seq · ChIP-seq · CUT&RUN · spatial transcriptomics · gene expression analysis · sequence alignment · epigenomic regulatory networks
- **ML & Algorithm Development:** XGBoost · PyTorch · scikit-learn · classification · feature engineering · model validation · deep learning · LLM output evaluation and quality control
- **Data Engineering & Curation:** SQL/MySQL · ETL pipelines · data validation and QC · biocuration workflows · scientific database development · metadata standardization · data integrity and provenance tracking
- **Programming & Infrastructure:** Python · R · SQL · Bash · Snakemake · Conda · Git/GitHub · Linux/Unix HPC · RShiny · JavaScript/AJAX · cloud computing

Professional Experience

Researcher · HIVE Lab, George Washington University *Jun 2025 – Present*

- Contributing to **GlycoSiteMiner**, an NLP and LLM-based bioinformatics pipeline that mines experimentally confirmed protein glycosylation sites from PubMed literature; curate and validate LLM-extracted candidates against published experimental evidence and enter records through GlyTableMaker.
- Apply glycobiology domain knowledge to distinguish true glycosylation events from common decoys (phosphorylation, glycation, unoccupied N-glycan sequons) and classify findings by evidence type and PTM origin.
- Authored a **structured feedback report** on the curation web interface covering accession validation, flagging logic, conditional workflows, and UX improvements.
- Contributing to **ML pipeline v2.0 development**: human-curated negative labeling, gold evaluation set construction, active learning, and richer feature engineering; roadmap toward extending pipeline coverage to additional post-translational modifications.

STEAM Instructor · iCode Columbia *Oct 2025 – Present*

- Design and deliver structured curricula on programming and debugging to students with varied technical backgrounds; communicate complex technical concepts clearly to non-specialist audiences.

Data Analyst (AEOP Fellow) · Walter Reed Army Institute of Research *Feb 2025 – Sep 2025*

- Developed and maintained **multi-omics analysis pipelines** (transcriptomics, metabolomics, lipidomics, microbiome) on Linux HPC clusters; applied statistical and ML approaches to derive insights from large clinical cohort datasets.
- Collaborated cross-functionally with bench scientists, clinicians, and PIs to interpret complex biological data and communicate findings to technical and non-technical stakeholders.

Image Analysis Intern · Carnegie Mellon University *Oct 2024 – Dec 2024*

- Developed **PyTorch deep learning algorithms** for large-scale cryoET imaging data analysis; contributed to algorithm optimization and scientific documentation for a translational drug discovery project.

Cancer Research Intern (CRI-SP Fellow) · National Cancer Institute, NIH *May 2024 – Aug 2024*

- Developed and optimized an **XGBoost classifier** on TCGA and ICGC high-throughput RNA-seq datasets to identify transcriptional phenocopies of the $KRAS^{G12D}$ mutation; engineered features, validated performance (ROC-AUC, cross-validation), and built the workflow as a one-command reproducible pipeline.
- Validated the algorithm on **single-cell RNA-seq** data and applied **spatial transcriptomics** to characterize tumor microenvironments; presented at NIH Summer Poster Day 2024.

Database Development Intern · Boston University *Jan 2024 – May 2024*

- Built the **cEpiReg Database**: integrated bioinformatics pipelines for scRNA-seq, snATAC-seq, ATAC-seq, and ChIP-seq (Seurat, ArchR, Scanpy); developed a cell-type-specific **TF–CRE–gene regulatory network framework**.
- Designed and maintained a **full-stack data platform** (RShiny, SQL, JavaScript/AJAX); organized version-controlled code repositories on GitHub; supported two peer-reviewed publications.

Education

Master of Science, Bioinformatics *Boston, MA · 2025*
Boston University

Bachelor of Science with Honors, Biotechnology *Bangalore, India · 2022*
M.S. Ramaiah University of Applied Sciences